MILITARY SPECIFICATION

CONNECTOR, ELASTIC TUBING, BRANCHED, Plastic, Y-Shaped

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers plastic Y-connectors for use with flexible tubing in laboratory or medical procedures.

1.2 Classification. Connectors shall be the following types and sizes as specified for the appropriate stock number and item identification:

Type I - with rings of varying diameter

Size 1 - for 1/4-inch tubing
Size 2 - for 3/8-inch tubing

Type II - with rings of same diameter

Size 1 - for 1/4 inch tubing
Size 2 - for 3/8-inch tubing

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Wood, Wirebound.
PPP-B-585	Boxes, Folding, Paperboard.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Shipping, Fiberboard.
PPP-B-676	Boxes, Setup.

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Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U.S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 19082.)

3. REQUIREMENTS

3.1 Material. Connectors shall be fabricated from plastic conforming to the requirements specified herein.

3.1.1 Chemical resistance. Connectors shall show no distortion, shall not become brittle, tacky or cloudy, and shall not be adversely affected when tested as specified in 4.4.1.

3.1.2 Heat resistance. Connectors shall show no distortion, shall not become brittle, tacky or cloudy, and shall not be adversely affected when tested as specified in 4.4.2.

3.1.3 Sterilization compatibility. Connectors shall show no distortion, shall not become brittle, tacky or cloudy, and shall not be adversely affected when tested as specified in 4.4.3.

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3.1.4 Toxicity. Connectors shall be non-toxic, when tested as specified in 4.4.4.

3.1.5 Heavy metals. Connectors shall not contain more than 10 parts per million (ppm.) of extractable heavy metals (as lead), when tested as specified in 4.4.5.

3.1.6 Resistance to shattering. Connectors shall not shatter, when tested as specified in 4.4.6.

3.1.7 Transparency. The flow of water shall be visible through the connector when tested as specified in 4.4.7.

3.2 Construction. Connectors shall be Y-shaped with all legs of equal length. Each leg shall be circled by three raised rings. Distance between rings and distance from the end of the leg to the outside ring shall be equal. For Type I, outside diameter of rings shall be progressively greater as the center of the Y is approached. For Type II, each ring shall have the same outside diameter.

3.3 Dimensions. Dimensions shall be as shown in Table I.

Table I Dimensions (in inches)

	Type	Size 1	Size 2
Length of legs	I	1.25 ± 0.06	1.56 ± 0.12
	II	1.06 ± 0.06	1.25 ± 0.06
Outside diameter of ring closest to the middle of the Y	I	0.319 ± 0.010	0.440 ± .015
Outside diameter of middle ring	I	0.295 ± 0.010	0.420 ± 0.010
Outside diameter of ring closest to end of leg	I	0.273 ± 0.010	0.390 ± 0.010
Outside diameter of each ring	II	0.31 ± 0.010	0.45 ± 0.010
Distance from end of leg to ring closest to the center of Y	I	0.75 ± 0.015	0.75 ± 0.015
	II	0.41 ± 0.010	0.48 ± 0.010
Wall thickness	I		
	II	0.018 min.	0.018 min.
Bore	I	0.18 (nom.)	0.3 nom.
	II	0.16 (nom.)	0.3 nom.

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3.4 Workmanship. Workmanship shall be first class throughout. Connectors shall be free from defects listed in Table IV and other defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table II. Unit of product shall be one connector.

Table II Sampling for examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
For dimensional examination	S-1	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table III. Unit of product shall be one connector, and acceptance number shall be zero.

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Table III Sampling for tests

Characteristic	Requirement	Test procedure	Inspection level
Chemical resistance	3.1.1	4.4.1	S-2 *
Heat resistance	3.1.2	4.4.2	S-1
Sterilization compatibility	3.1.3	4.4.3	S-2 *
Toxicity	3.1.4	4.4.4	S-1
Heavy metals	3.1.5	4.4.5	S-1
Resistance to shattering	3.1.6	4.4.6	S-1
Transparency	3.1.7	4.4.7	S-1

* The same number of connectors shall be used for each part of the test, and the aggregate shall equal the amount indicated. If necessary, extra connectors shall be sampled to achieve this equality.

4.3 Examination. The connectors shall be examined to determine compliance with all requirements contained in this specification. Examination shall be in accordance with, but shall not be restricted to, the defects listed in Table IV.

Table IV Classification of defects

Category	Defect
Major	
101	Connector not free of cracks.
102	Connector not free of deformity.
103	Connector not free of blisters.
104	Connector not free of grease or other contamination.
105	Connector not free of tackiness.
106	Connector not free of flash.

4.3.1 Dimensional examination. The connectors shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Chemical resistance. Connectors shall be immersed in the following solutions and liquids for 24 hours at the indicated temperature.

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Solution or solvent	Temperature *
30% Sulfuric Acid (aqueous)	40° C.
10% Sodium Hydroxide (aqueous)	40° C.
95% Ethyl Alcohol (aqueous)	35° C.
Toluene	25° C.
Acetone	25° C.

* ± 2° shall be permitted.

4.4.2 Heat resistance. The connectors shall be subjected to a temperature of 250° ± 2° Fahrenheit (°F.) for one hour.

4.4.3 Sterilization compatibility. Connectors shall be tested in accordance with the following procedures:

a. Total immersion in an aqueous solution (1:750) of Benzalkonium Chloride U.S.P. for 24 hours at room temperature
b. Total immersion in boiling water for 30 minutes (min.)
c. Autoclaving (15 pounds per square inch (psi) pressure, 121°C. for 30 minutes).

4.4.4 Toxicity. Connectors shall be tested for toxicity by the following methods:

Break connectors into small segments. Place the segments in 100 milliliters (ml.) of Sodium Chloride Injection U.S.P. and autoclave the mixture for 1 hour at 250° F. Inject the saline extract into each of five mice weighing 18-25 grams intravenously via the tail vein at a dose level of 50 ml./kg. Examine mice for evidence of toxicity.

4.4.5 Heavy metals. The ashed residue of the 10% nitric acid extract from 10 grams of plastic connectors shall be tested as specified in 4.4.5.1.

4.4.5.1 Test for lead. Ten grams of sample shall be taken, ignited to char, and then further ignited in a muffle at 700° ± 5° C. to a white ash. The ash shall be boiled in 15 ml. of 10% nitric acid, filtered into a volumetric flask and made up to 100 ml. An aliquot, equivalent to one gram of sample, shall be taken and placed in a separator. The dithizone test for lead, as given in the current edition of the U.S.P. shall be applied, using a control of 5 ppm. or 12 ppm. of lead solution, as given in the U.S.P. test. If sample solution is darker than the limit, smaller aliquots shall be used until the color of the control is matched by the sample solution and the concentration of lead in ppm. is determined.

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4.4.6 Resistance to shattering. Cut through the leg of the connector with a sharp instrument.

4.4.7 Transparency. Pass water through all three legs of the connector.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One package, containing 50 connectors, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table V shall be packaged in each unit, intermediate and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table V Packaging quantities

Unit package	Intermediate package	Exterior container
1 unit	10 units	100 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be a box of appropriate size and design constructed in accordance with PPP-B-566 or PPP-B-676, except commercial colors will be acceptable, or PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent accidental opening under normal handling.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-676, except that commercial colors will be acceptable; or PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent accidental opening under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

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5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof in accordance with 30.4 of PPP-B-636.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be closed by method II of PPP-B-636. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Applicable size and type required (see 1.2).
d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification covers the following items as listed in the Federal Supply Catalog. Particular sizes may be procured under the applicable NSN.

Size	National Stock No.	Item Identification
1	6640-00-725-7843	CONNECTOR, PLASTIC TUBING, BRANCHED, Plastic, Y-Shaped, 5/16 inch Outside Diameter, 50s
2	6640-00-900-7133	CONNECTOR, PLASTIC TUBING, BRANCHED, Plastic, Y-Shaped, 7/16 inch Outside Diameter, 50s

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6640-0840
CS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS. The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-C-37068

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
☐ YES ☐ NO (If "Yes", in what way?)

4. REMARKS

SUBMITTED BY (Printed or typed name and address - optional)

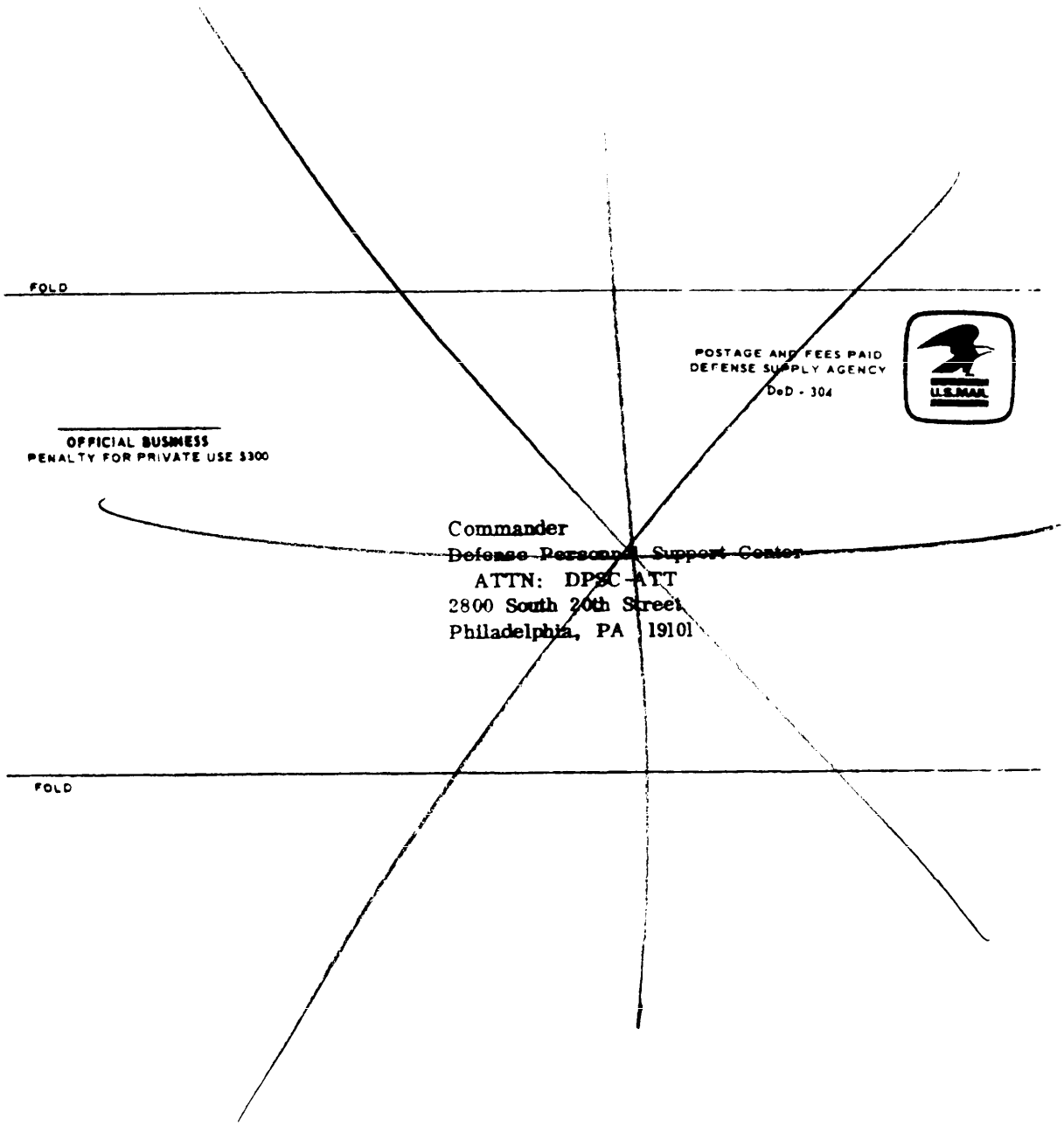
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DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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